

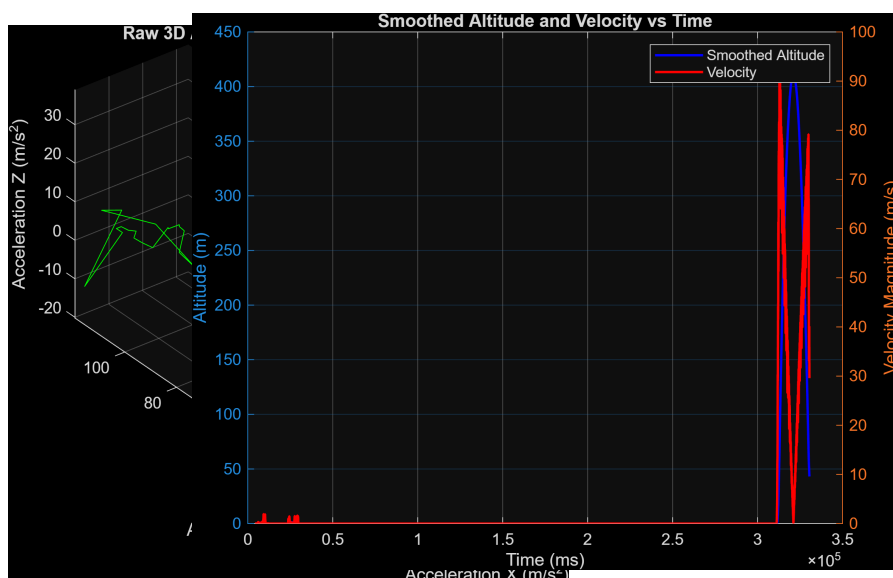
Flight Summary - ARCS Team 1 - Electric Dysfunction

The rocket demonstrated an aggressive launch profile that was ultimately compromised by a mechanical recovery failure. Off the pad, the vehicle exhibited notable angular deviation, characterized by sudden pitch and yaw oscillations captured in the raw gyroscope data and visually. This behavior is likely due to an overstable configuration (simulated at 2.05 calibers). After clearing the launch rail, the rocket instantly weathercocked into the crosswind, corrected its path, but overcorrected resulting in some oscillations before locking into a stable upwards trajectory. Our measured apogee was 411m, which cannot be confirmed with the Committee Research Payload as the data was lost when the power was disconnected. This less than expected apogee can likely be put down to the initial oscillations increasing drag and cosine losses.

The primary failure mode occurred at apogee, where the airframe failed to decouple. The late addition of recovery wadding and a tightly packed Nomex protector sheet restricted the linear volume inside the airframe. Under the pressure of the ejection charge, this tightly packed bundle likely compressed and expanded radially, jamming itself against the inner walls of the recovery bay. Because the gas could not bypass or push through this fabric plug to decouple, the ejection gases forced its way backward and blew out through the spent motor casing instead of separating the airframe. Lacking parachute deployment, the vehicle entered a stable ballistic descent, maintaining structural integrity until it impacted the ground at 80 m/s.

After searching through the tall grass and wild field for around 2 hours, we eventually recovered the rocket. The PETG nose cone was buried in the ground and had sheared off, whilst the forces travelled through the stainless steel rods in the AV bay, snapping the rocket in half, subsequently obliterating the fin assembly. After prying the AV bay out, we were pleased to discover that the AV bay had drastically overperformed structurally and was powered on and logging. Despite the 180 mph impact, the 3D printed PETG sled, mounting infrastructure and solder joints remained intact and flight logging data uncorrupted. This was a huge success in terms of the useful data we acquired and as such the rocket was not a total failure.

MATLAB Data Plots



80 m/s impact

